geometry, wasn't sure, but after browsing in the library he came back with an answer: Yes, the necessary equations did exist. They had been invented largely by the German mathematician Bernhard Riemann in the 1860s, the Italian Gregorio Ricci in the 1880s, and Ricci's student Tullio Levi-Civita in the 1890s and 1900s; they were called the "absolute differential calculus" (or, in physicists' language of 1915–1960, "tensor analysis," or in the language of 1960 to the present, "differential geometry"). But, Grossmann told Einstein, this differential geometry is a terrible mess which physicists should not be involved with. Were there any other geometries that could be used to figure out the law of warpage? No.

And so, with much help from Grossmann, Einstein set out to master the intricacies of differential geometry. As Grossmann taught mathematics to Einstein, Einstein taught something of physics to Grossmann. Einstein later quoted Grossmann as saying, "I concede that I did after all gain something rather important from the study of physics. Before, when I sat on a chair and felt a trace of heat left by my 'pre-sitter,' I used to shudder a little. That is completely gone, for on this point physics has taught me that heat is something completely impersonal."

Learning differential geometry was not an easy task for Einstein. The spirit of the subject was alien to the intuitive physical arguments that he found so natural. In late October 1912 he wrote to Arnold Sommerfeld, a leading German physicist: "I am now occupying myself exclusively with the problem of gravitation and believe that, with the aid of a local mathematician [Grossmann] who is a friend of mine I'll now be able to master all the difficulties. But one thing is certain, that in all my life I have never struggled so hard, and that I have been infused with great respect for mathematics the subtler parts of which, in my simple-mindedness, I had considered pure luxury up to now! Compared to this problem the original relativity theory [special relativity] is child's play."

Together Einstein and Grossmann struggled through the autumn and into the winter with the puzzle of how matter forces spacetime to curve. But despite their all-out effort, the mathematics could not be brought into accord with Einstein's vision. The law of warpage eluded them.

Einstein was convinced that the law of warpage should obey a *generalized (enlarged) version of his principle of relativity:* It should look the same in every reference frame—not just inertial (freely falling) frames, but non-inertial frames as well. The law of warpage should not rely for its